QR-003 Wet layup and the bag stack

Doc ID	QR-003
Title	Wet layup and the bag stack
Parent	CS-007 §7.1; CS-007 §7.2; CS-007 §7.3; CS-007 §7.4; CS-007 §7.5; CS-007 §7.6; CS-007 §7.7
Checked against	CS-007 Rev D; CS-007 Rev D; CS-007 Rev D; CS-007 Rev D; CS-007 Rev D; CS-007 Rev D; CS-007 Rev D

Derived from its parent standard. If this and the parent disagree, the parent is right.

1. Wet layup bag stack, in order: peel ply, breather, bag

From the part up: wet laminate, peel ply past the laminate edges, breather over the peel ply but held back from every part edge, then the VB160 bag sealed with tacky tape, with a resin trap between bag and pump (CS-007 §7.4).

Also called: bag stack, bagging order, vacuum bag, bag, breather, bleeder, batting, peel ply, release ply, wet layup bag.

1. Lay peel ply over the full laminate, past its edges (CS-007 §7.4 step 1).
2. Lay breather over the peel ply, held back from every part edge it could bond to (CS-007 §7.4 step 2).
3. Run tacky tape and bag with VB160 (CS-007 §7.4 step 3).
4. Pleat every corner, and every 300–400 mm of a curved perimeter (CS-007 §7.4 step 3).
5. Put the resin trap between bag and pump, per CS-006 §7.3.

How far back to hold the breather: CS-007 sets no distance. The rule is that it touches no part edge it could bond to, because resin that wicks into breather at an edge glues it on (CS-007 §4).

Where the work order says so, you may swap the peel ply for a release film (A4000R) or a mylar sheet, or add a caul plate over the peel ply. These are still under test (2026-09): write which one you used on the work order and photograph the finished surface (CS-007 §7.4 step 4).

2. Before you mix West: shop temperature and what to cut first

Check the shop is at or above the hardener's minimum before mixing: 206 needs 60 °F (16 °C), 205 needs 40 °F (4 °C), 209 needs 70 °F (21 °C). These are the West datasheet minimums (CS-007 §7.1 step 4).

Also called: minimum temperature, cold shop, hardener choice, 205, 206, 209, prep, wet layup prerequisites.

Below the minimum, pick a hardener whose minimum the shop meets, or wait.

1. Check the stack is frozen and printed on the work order (CS-007 §7.1 step 1).
2. Check the mold is sealed and released per CS-004 with its checks passed, or the plate is waxed, or the foam core is shaped and dry-fitted.
3. Cut the cloth per the work order's ply table (CS-007 §7.1 step 2).
4. Weigh the dry cloth and write the total dry fiber mass on the work order.
5. Cut peel ply, breather, bag and tacky tape to size before any resin is mixed (CS-007 §7.1 step 3). Pot life spent hunting scissors is pot life lost.
6. Read the shop temperature and check it against the hardener (CS-007 §7.1 step 4).

3. How much West resin to mix, and the pump ratio

Mix about dry fiber mass × 0.9 of resin. Pump 5:1 by volume for 206 and 205, one full stroke of each pump per shot, never partial strokes. 209 is 3:1 on a different pump head (CS-007 §7.2).

Also called: resin math, how much resin, West, West System, 105, pumps, mix ratio, how much hardener.

The 0.9 is more than the 60:40 fiber to resin target implies, on purpose: the squeegee and breather take the excess out, and the cured part lands near 60:40 (CS-007 §7.2 step 1).

Hardener	Pumps	By scale (weight)	Pot life per 100 g at 72 °F
206 slow (default)	5:1	5.36:1	20–25 min
205 fast	5:1	5.19:1	9–12 min
209 extra slow	3:1, different pump head	3.68:1	40–50 min

Sources: West TDSs, as copied into CS-007 §7.2.

1. Work out the mix: dry fiber mass × 0.9.
2. Check the pump head matches the hardener before the first stroke.
3. Pump full strokes only, one of each per shot.
4. Mix per CS-008: at least 2 min, then double-pot.
5. Write mixed and final masses on the work order (CS-007 §7.2 step 1).

A bigger job is several small pots, not one large one. A pot that warms is about to gel (CS-007 §7.2 step 3).

4. Wetting out each ply

Wet out ply by ply: resin on, then squeegee until the ply is shiny-wet with the weave still visible, not swimming. Chase every dry, white patch before the next ply goes on; you will not see it again (CS-007 §7.3 step 1).

Also called: wet out, wet-out, wetting, squeegee, hand layup, dry patch, white fibre, white fiber.

1. Put resin on the ply.
2. Squeegee until it is shiny-wet with the weave visible.
3. Look for dry, white patches and work resin into them.
4. Check the ply against the run's stack ("Stack & BOM") as it goes on (CS-007 §7.3 step 2).
5. Put on the next ply.

A substitution or an extra ply gets edited into the run's stack, with the reason in that ply's Notes (CS-007 §7.3 step 2). A pen tick on a printed traveler is allowed as well.

Swimming in resin is past wet-out, not better (CS-007 §4). The acceptance check is no white or dry weave visible, ply by ply (CS-007 §8).

5. Vacuum and drop test for a wet layup

Pull to 25 inHg or better, then isolate the pump: the bag shall lose 2 inHg or less over 10 minutes, then open the valve again. Write the start and end readings in the note on "Wet layup and bag" ("Wet layup over core" on a Foam wrapped run) (CS-007 §7.5).

Also called: drop test, leak test, leak-down, vacuum level, inHg, 25 inHg, vacuum crush.

1. Pull to 25 inHg or better (CS-007 §7.5 step 1).
2. Close the valve to isolate the pump. The gauge must be on the bag side of the closed valve (CS-007 §7.5 step 2).
3. Read the gauge, wait 10 minutes, read it again.
4. Pass is a loss of 2 inHg or less (CS-007 §7.5 step 2).
5. Write both readings in the step note.
6. Failed? Hunt leaks per CS-006 §7.7 (QR-012) and test again.
7. Passed? Open the valve again and check the gauge climbs back to 25 inHg before you walk away (CS-007 §7.5 step 4).

On light foam cores, watch for vacuum crush. If the core dents, reduce the vacuum and write the level that held on the work order, per part (CS-007 §7.5 step 3).

There is no separate drop-test step on the Mold wet lay or Foam wrapped techniques, so the app does not block on this test. A drop-test step placed before "Cure and demould" or "Cure" would switch the cure hold off, which is why the readings go in the "Wet layup and bag" note (CS-007 §7.5).

6. Signing the layup and when to demold (demould)

Leave it bagged under vacuum overnight at least. The FEB hold before demold is 24 h for 206 and 205 and 36 h for 209, counted from when you finished bagging (CS-007 §7.6). These are team holds; the datasheet thin-film cure is 10–15 h, 6–8 h and 20–24 h at 72 °F.

Also called: demold, demould, de-mould, cure hold, FEB hold, sign off, buy-off, cure time, pull the part, when can I pull it, take it off the mold, how long to wait, unbag.

1. Leave the part bagged under vacuum overnight at least (CS-007 §7.6 step 1).
2. Before you leave, sign "Wet layup and bag" ("Wet layup over core" on a Foam wrapped run). Signing needs Wet layup training.
3. Under "Resin mixed", pick West 105 + 206 slow, West 105 + 205 fast or West 105 + 209 extra slow, whichever you mixed.
4. Set "Finished on" and "at" to when you finished bagging.
5. Fill in "Shop temperature, °C (optional)". This starts the FEB hold.
6. Come back after the hold. "Cure and demould" ("Cure" on Foam wrapped) stays locked until then (CS-007 §7.6 step 2). Only a lead can sign it early, and the app makes them write why.
7. Write times and masses on the work order, label the part per CS-001 and hand it on for trim (CS-007 §7.6 step 3).
8. On the Molds tab, open the mold, add 1 to Parts pulled and set Stage back to Sealed: the PVA came off with the part (CS-007 §7.6 step 4).

A cold shop runs longer than these figures (CS-007 §7.6 step 1). Every resin's hold and why: QR-001.

7. Foam-wrapped wing elements (Foam wrapped technique)

The same wet layup over a shaped foam core instead of a mold: wet plies wrap the core, then peel ply, breather, bag. Watch for vacuum crush on the core: if it dents, reduce vacuum and write the level that held (CS-007 §7.7; §7.5 step 3).

Also called: foam core, foam wrapped, foam-wrapped, wing element, RW elements, E2 to E5, core.

In the app this is the Foam wrapped technique. Its steps are (CS-007 §7.7):

1. "Stack frozen".
2. "Shape foam core" (needs Foam core training).
3. "Wet layup over core" (needs Wet layup training). Signing it starts the cure hold.
4. "Cure" (the hold).
5. "Trim and finish".

Everything else, the resin math, wet-out, drop test and hold, is the same as a wet layup on a mold (QR-003 §3 to §6).